

A Hopf algebra on nonplanar binary forests

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Linguistic foundations

The fundamental objects in Marcolli, Chomsky and Berwick's model of Merge are **workspaces of syntactic objects**:

- nonplanar binary forests with leaves labelled by **lexical items**

Merge in its simplest form takes two binary trees and combines them together at a common root.

$$\mathfrak{M}(T, S) = \begin{array}{c} \diagup \quad \diagdown \\ T \quad S \end{array} = \begin{array}{c} \diagup \quad \diagdown \\ S \quad T \end{array} = \mathfrak{M}(S, T)$$

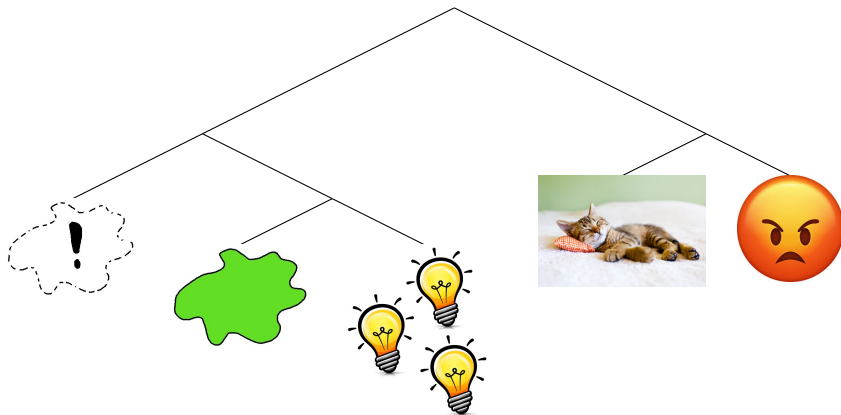
it is commutative but not associative

$$\mathfrak{M}(T, \mathfrak{M}(S, R)) = \begin{array}{c} \diagup \quad \diagdown \\ T \quad \begin{array}{c} \diagup \quad \diagdown \\ S \quad R \end{array} \end{array} \neq \begin{array}{c} \diagup \quad \diagdown \\ \begin{array}{c} \diagup \quad \diagdown \\ T \quad S \end{array} \quad R \end{array} = \mathfrak{M}(\mathfrak{M}(T, S), R)$$

Why nonplanar?

- Word order varies across human languages.
- Yet it cannot vary arbitrarily; studies show that certain syntactic rules do not naturally appear in human languages.

For your edification (IYKYK)



Hopf algebras

A **Hopf algebra** $\mathcal{H}$ is a bialgebra

- algebra (multiplication, or product, $m : \mathcal{H} \otimes \mathcal{H} \rightarrow \mathcal{H}$)
- coalgebra (decomposition, or coproduct, $\Delta : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{H}$)

with a unique antipode map $S : \mathcal{H} \rightarrow \mathcal{H}$.

We assume that

- m is associative:

$$m(m(x, y), z) = m(x, m(y, z))$$

i.e., $(x \cdot y) \cdot z = x \cdot (y \cdot z)$.

- Δ is coassociative:

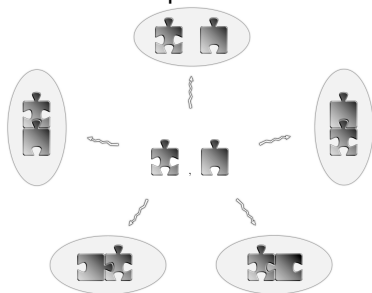
$$(\Delta \otimes \mathbb{1})\Delta = (\mathbb{1} \otimes \Delta)\Delta.$$

Combinatorial Hopf algebras

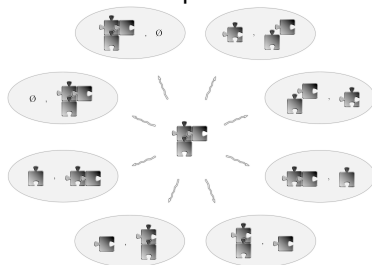
Many Hopf algebras (including ours) are combinatorial in nature:

- the product is indexed by “ways to *put things together*”
- the coproduct is indexed by “ways to *break things apart*”.

Composition:



Decomposition:



Figures courtesy of:

Pawel Blasiak. “Combinatorial Route to Algebra: The Art of Composition & Decomposition”. arXiv:1008.4685 [math.CO]

Graded Hopf algebras

A combinatorial Hopf algebra is often **graded**, meaning that the underlying vector space $\mathcal{H} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ is graded, and $\mathcal{H}$ is

- graded as an algebra:

$$m(\mathcal{H}_k, \mathcal{H}_\ell) \subseteq \mathcal{H}_{k+\ell};$$

- graded as a coalgebra:

$$\Delta(\mathcal{H}_n) \subseteq \bigoplus_{k=0}^n \mathcal{H}_k \otimes \mathcal{H}_{n-k}.$$

If $\mathcal{H}$ is graded, then it is furthermore

- **connected** if $\dim \mathcal{H}_0 = 1$;
- **locally finite** if $\dim \mathcal{H}_k < \infty$ for any k .

A graded connected bialgebra is *automatically* a Hopf algebra as it has a recursively-defined antipode.

Examples of Hopf algebras

Example

Polynomial Hopf algebra $(k[X], m, \Delta)$; m is usual product and

$$\Delta(X) = 1 \otimes X + X \otimes 1, \quad \Delta(X^n) = \sum_{i=0}^n \binom{n}{i} X^i \otimes X^{n-i}.$$

It is commutative and cocommutative.

Example

Hopf algebra of words $(\mathcal{W}, \cdot, \text{III})$ with concatenation product and unshuffle coproduct (cocommutative but *not* commutative):

$$abc \cdot de = abcde$$

$$\begin{aligned} \text{III}(abc) = & 1 \otimes abc + abc \otimes 1 + a \otimes bc + b \otimes ac + c \otimes ab \\ & + bc \otimes a + ac \otimes b + ab \otimes c. \end{aligned}$$

A coproduct for binary forests

Now we introduce a coproduct for the space of binary, rooted, nonplanar forests, for example:

$$T = \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}, \quad F = \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \quad \delta \end{array} \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \epsilon \quad \zeta \end{array}$$

Definition

A **binary-admissible cut** of T can be written as a subset $C \subseteq E(T)$ of edges such that

- ① no pair of edges in C are on the same path from root to leaf,
- ② no two edges in C share the same parent vertex.

Conversely, any such subset of $E(T)$ defines a binary-admissible cut.

A Hopf algebra on nonplanar binary trees

- $\mathcal{H} = \mathcal{H}_{bin}$ is the algebra generated by nonplanar rooted binary trees (labelled or unlabelled), with disjoint union as product.
- Δ is the coproduct based on binary-admissible cuts and edge contraction.

Example

$$\begin{aligned} \Delta\left(\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}\right) = & 1 \otimes \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array} + \alpha \otimes \begin{array}{c} \diagup \quad \diagdown \\ \beta \quad \gamma \end{array} + \beta \otimes \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \gamma \end{array} \\ & + \gamma \otimes \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} + \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \end{array} \otimes \gamma + \alpha \sqcup \gamma \otimes \beta \\ & + \beta \sqcup \gamma \otimes \alpha + \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array} \otimes 1 \end{aligned}$$

Theorem

Δ is coassociative.

Another interpretation of the coproduct

We can also index the terms of the coproduct by subsets of $L(T)$:

Example

$$\begin{aligned} \Delta(\text{tree with root } \alpha \text{ and children } \beta, \gamma) = & 1 \otimes \text{tree} + \alpha \otimes \text{tree} + \beta \otimes \text{tree} + \gamma \otimes \text{tree} \\ & + \alpha \sqcup \gamma \otimes \beta \\ & + \beta \sqcup \gamma \otimes \alpha + \text{tree} \otimes 1 \end{aligned}$$

Here the terms correspond to

$$\emptyset, \{\alpha\}, \{\beta\}, \{\gamma\}, \{\alpha, \beta\}, \{\alpha, \gamma\}, \{\beta, \gamma\}, \{\alpha, \beta, \gamma\}.$$

with trivial subsets $\emptyset, L(T)$ matching the primitive terms.

$\mathcal{H}$ is graded by the number of leaves and connected.

The antipode

Since $\mathcal{H}$ is graded connected, its **antipode** is recursively defined.

Example

$$\begin{aligned}
 S(\alpha) &= -\alpha \\
 S(\alpha \frown \beta) &= -\alpha \frown \beta + 2\alpha \sqcup \beta \\
 S(\alpha \frown \beta \frown \gamma) &= -\alpha \frown \beta \frown \gamma + \alpha \sqcup \beta \frown \gamma + \beta \sqcup \alpha \frown \gamma \\
 &\quad + 2\gamma \sqcup \alpha \frown \beta - 4\alpha \sqcup \beta \sqcup \gamma.
 \end{aligned}$$

- terms in $S(T)$ correspond to the **binary-total cuts** of T .
- if T has n leaves, the unsigned sum of all coefficients in $S(T)$ is 3^{n-1} .
- the sign of a term in $S(T)$ is negative if it has an odd number of trees; it is positive if it has an even number of trees.

Binary-total cuts

We enumerate the binary-total cuts of $T = \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$:

binary-total cut	term in $S(T)$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$ (empty cut)
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\gamma \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \end{array}$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\gamma \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \end{array}$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\alpha \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \beta \quad \gamma \end{array}$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\beta \sqcup \begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \gamma \end{array}$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\alpha \sqcup \beta \sqcup \gamma$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\alpha \sqcup \beta \sqcup \gamma$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\alpha \sqcup \beta \sqcup \gamma$
$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$	$\alpha \sqcup \beta \sqcup \gamma$

Lie algebras

Definition

A **Lie algebra** is a vector space V with a bilinear product $[a, b]$ satisfying the Lie axioms:

- antisymmetry:

$$[a, b] = -[b, a]$$

- Jacobi identity:

$$[a, [b, c]] + [b, [c, a]] + [c, [a, b]] = 0.$$

Such a product $[a, b]$ is called the **Lie bracket**.

Pre-Lie algebras

Let V be a vector space, $\triangleleft: V \times V \rightarrow V$ a bilinear product. The associator of $\triangleleft$ is

$$A_{\triangleleft}(a, b, c) = (a \triangleleft b) \triangleleft c - a \triangleleft (b \triangleleft c)$$

Definition

We call $\triangleleft$ a **pre-Lie operator** if it satisfies the **Vinberg identity**

$$A_{\triangleleft}(a, b, c) = A_{\triangleleft}(a, c, b)$$

making $(V, \triangleleft)$ a **pre-Lie algebra**.

- A pre-Lie operator induces a Lie bracket defined by $[a, b] := a \triangleleft b - b \triangleleft a$, which automatically satisfies the Lie axioms, making V a Lie algebra.

⚠ However, not all Lie algebras come from a pre-Lie operator!

Definition

$T \triangleleft S$ is the sum of the trees produced by inserting S at each edge of T , **plus** the tree obtained by grafting T and S to a common root (which is $\mathfrak{M}(T, S)$, the result of applying Merge to T and S).

This will be our pre-Lie operation.

Example

$$\begin{array}{c}
 \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \gamma \triangleleft \begin{array}{c} \diagup \\ \delta \end{array} \begin{array}{c} \diagdown \\ \epsilon \end{array} = \\
 \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \begin{array}{c} \diagup \\ \delta \end{array} \begin{array}{c} \diagdown \\ \epsilon \end{array} \gamma + \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \begin{array}{c} \diagup \\ \delta \end{array} \begin{array}{c} \diagdown \\ \epsilon \end{array} \gamma \\
 + \\
 \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \delta \end{array} \begin{array}{c} \diagup \\ \epsilon \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \gamma + \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \delta \end{array} \begin{array}{c} \diagup \\ \epsilon \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \gamma \\
 + \\
 \begin{array}{c} \diagup \\ \alpha \end{array} \begin{array}{c} \diagdown \\ \beta \end{array} \gamma \begin{array}{c} \diagup \\ \delta \end{array} \begin{array}{c} \diagdown \\ \epsilon \end{array}
 \end{array}$$

If you are concerned by the idea of “gratuitously” including the Merge term, you can pretend that each tree has a “ghost” edge at the root:

$$\begin{array}{c} e \\ | \\ \alpha \quad \beta \quad \gamma \end{array} \triangleleft_e \begin{array}{c} | \\ \delta \quad \epsilon \end{array} = \begin{array}{c} | \\ \alpha \quad \beta \quad \gamma \quad \delta \quad \epsilon \end{array}$$

and let the insertion occur over all edges.

From a pre-Lie algebra to a(nother) Hopf algebra

The **universal enveloping algebra** of a Lie algebra constructed from a pre-Lie operation is a Hopf algebra. A construction of $\mathcal{U}(\mathcal{L})$, as given by Oudom-Guin:

- ① Identify $\mathcal{U}(\mathcal{L})$ with $\text{Sym}(\mathcal{L})$ (polynomials over basis of $\mathcal{L}$)
- ② Extend $\triangleleft$ from $\mathcal{L}$ to $\text{Sym}(\mathcal{L})$
- ③ Modify $\triangleleft$ to $* = *_{\triangleleft} : \text{Sym}(\mathcal{L}) \rightarrow \text{Sym}(\mathcal{L})$; this will be the product.
- ④ Equip it with the shuffle coproduct of $\text{Sym}(\mathcal{L})$:

$$\text{III}(x_1 x_2 \dots x_n) = \sum_{I \subseteq [n]} x_I \otimes x_{[n] \setminus I}.$$

$(\text{Sym}(\mathcal{L}), *, \text{III})$ is isomorphic to $\mathcal{U}(\mathcal{L})$. We will just call it $\mathcal{U}(\mathcal{L})$ since this is the only presentation we use.

$\mathcal{U}(\mathcal{L})$ is noncommutative and cocommutative.

Connecting the two Hopf algebras

In each of the previous sections, we introduced a Hopf algebra:

- $(\mathcal{H}, \sqcup, \Delta)$ - the Hopf algebra of edge-cutting on binary trees.
- $(\mathcal{U}(\mathcal{L}), *, \text{III})$ - the universal enveloping algebra of the Lie algebra of edge-insertion.

It seems natural that edge-cutting and edge-insertion should be “opposite” operations. In fact,

- edge-cutting deletes an edge and then contracts two edges into one, removing a vertex
- edge-insertion splits an existing edge into two by adding a vertex, and then creates a new edge

We formalize this using the concept of **duality**.

Duality of Hopf algebras

- $(\mathcal{H}, m, \Delta)$ is a (generic) Hopf algebra.
- $\langle -, - \rangle : \mathcal{H} \times \mathcal{H}^\vee \rightarrow k$ is a pairing on the underlying vector space of $\mathcal{H}$ (we identify the vec spaces underlying $\mathcal{H}$ and $\mathcal{H}^\vee$)
- The **dual of $\mathcal{H}$ with respect to $\langle -, - \rangle$** is the Hopf algebra $(\mathcal{H}^\vee, m^\vee, \Delta^\vee)$ defined by

$$\langle \Delta(x), y \otimes z \rangle = \langle x, m^\vee(y, z) \rangle$$

$$\langle m(x, y), z \rangle = \langle x \otimes y, \Delta^\vee(z) \rangle$$

The product and coproduct “swap places” in the dual,

Symmetries of forests

If F is a nonplanar binary forest, then we say

- $SG(F)$ is the group of **forest automorphisms on F** :
- $s_F = |SG(F)|$ is the cardinality of automorphism group.

If T is a binary tree, the only automorphisms of T swap the children of nodes with identical children.

Lemma

If T has r interior nodes with identical children, then $s_T = 2^r$.

Example

T	α	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \alpha \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \gamma \end{array}$
s_T	1	1	2	1
T	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \alpha \quad \beta \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \beta \quad \alpha \quad \beta \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \alpha \quad \beta \quad \beta \end{array}$	$\begin{array}{c} \diagup \quad \diagdown \\ \alpha \quad \alpha \quad \alpha \quad \alpha \end{array}$
s_T	2	2	4	8

Duality of $\mathcal{H}_{bin}$ and $\mathcal{U}(\mathcal{L})$

- $k[\mathcal{T}]$ is the polynomial algebra of nonplanar binary trees; it is the underlying vector space of both $\mathcal{H}$ and $\mathcal{U}(\mathcal{L})$.
- Define a pairing on $k[\mathcal{T}]$ by

$$\langle F, F' \rangle := s_F \delta_{F, F'}$$

for any forests $F, F' \in k[\mathcal{T}]$.

Theorem

*The Hopf algebra $(\mathcal{H}, \sqcup, \Delta)$ is dual to the **opposite** of $(\mathcal{U}(\mathcal{L}), *, \text{III})$ with respect to $\langle -, - \rangle$, in the sense that their (co)products are related by*

$$\begin{aligned} \langle \text{III}(F_1), F_2 \otimes F_3 \rangle &= \langle F_1, F_2 \sqcup F_3 \rangle \\ \langle F_1 * F_2, F_3 \rangle &= \langle F_2 \otimes F_1, \Delta(F_3) \rangle. \end{aligned}$$

First equality is “easy” exercise. Second one is a bit more involved.

Proving duality of $*$ and Δ

Key to the proof of duality: counting the multiplicities of cuts and grafts (the same strategy used by Hoffman to prove duality of the Butcher-Connes-Kreimer and Grossman-Larson Hopf algebras).

Definition

Let F_1, F_2, F_3 be forests. Define

$n(F_1, F_2; F_3)$ = the coefficient of F_3 in $F_2 * F_1$

$m(F_1, F_2; F_3)$ = the coefficient of $F_1 \otimes F_2$ in $\Delta(F_3)$.

The goal now is to show that

$$n(F_1, F_2; F_3) s_{F_3} = m(F_1, F_2; F_3) s_{F_1} s_{F_2}$$

by counting cut-classes (to determine $m(-, -; -)$) and graft-classes (to determine $n(-, -; -)$).

Example: Cut-classes and graft-classes

Let

$$F_1 = \wedge, F_2 = \bullet \bullet \wedge, F_3 = \begin{array}{c} \diagup \quad \diagdown \\ | \quad | \\ \diagdown \quad \diagup \\ | \quad | \\ \diagup \quad \diagdown \end{array},$$

which have symmetry coefficients

$$s_{F_1} = 2, s_{F_2} = 4, s_{F_3} = 4.$$

Since F_2 contains two copies of $\bullet$ and one copy of $\wedge$, there are $2! \cdot 1! = 2$ disjoint copies of $\mathcal{G}''$ in $\mathcal{G}'$. So $|\mathcal{C}'| = 7$ and $|\mathcal{G}'| = 2! \cdot |\mathcal{G}''| = 14$.

$$|\mathcal{C}'| s_{F_1} s_{F_2} = 7 \cdot 2 \cdot 4 = 56 = 14 \cdot 4 = |\mathcal{G}'| s_{F_3}.$$

Cut-classes in $\mathcal{C}'$	Graft-classes in $\mathcal{G}''$

Vielen Dank 😊

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